

3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

04:20

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i \text{ OR } B_i$).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Code:

```
import numpy as np
```

```
def array_operations(A, B):
```

```
    # Convert A and B to NumPy arrays
```

```
    A=np.array(A)
```

```
B=np.array(B)
```

```
# Arithmetic Operations
```

```
sum_result = A+B
```

```
diff_result = A-B
```

```
prod_result = A*B
```

```
# Statistical Operations
```

```
mean_A = np.mean(A)
```

```
median_A = np.median(A)
```

```
std_dev_A = np.std(A)
```

```
# Bitwise Operations
```

```
and_result = A&B
```

```
or_result = A|B
```

```
xor_result = A^B
```

```
# Output results with one space between each element
```

```
print("Element-wise Sum:", ' '.join(map(str, sum_result)))
```

```
print("Element-wise Difference:", ' '.join(map(str, diff_result)))
```

```
print("Element-wise Product:", ' '.join(map(str, prod_result)))
```

```
print(f"Mean of A: {mean_A}")
```

```
print(f"Median of A: {median_A}")
```

```
print(f"Standard Deviation of A: {std_dev_A}")
```

```

print("Bitwise AND:", ' '.join(map(str, and_result)))

print("Bitwise OR:", ' '.join(map(str, or_result)))

print("Bitwise XOR:", ' '.join(map(str, xor_result)))

```

```
A = list(map(int, input().split())) # Elements of array A
```

```
B = list(map(int, input().split())) # Elements of array B
```

```
array_operations(A, B)
```

Average time
0.039 s
39.00 ms

Maximum time
0.061 s
61.00 ms

✓ 1 out of 1 shown test case(s) passed
✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 61 ms Debug

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: -6·8·10·12	Element-wise Sum: -6·8·10·12
Element-wise Difference: -4·-4·-4·-4	Element-wise Difference: -4·-4·-4·-4
Element-wise Product: -5·12·21·32	Element-wise Product: -5·12·21·32
Mean of A: -2.5	Mean of A: -2.5
Median of A: -2.5	Median of A: -2.5
Standard Deviation of A: -1.118033988749895	Standard Deviation of A: -1.118033988749895
Bitwise AND: -1·2·3·0	Bitwise AND: -1·2·3·0
Bitwise OR: -5·6·7·12	Bitwise OR: -5·6·7·12
Bitwise XOR: -4·4·4·12	Bitwise XOR: -4·4·4·12